

Multilingual Hope Speech Detection

Low-Resource and Code-Mixed Settings

Detecting supportive and encouraging content in Tamil, Malayalam, and English

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What is Hope Speech? 🤔

Hope speech = Messages that comfort, encourage, and promote optimism

✓ Hope Speech Examples

- "I know this is hard, but you are not alone"
- "Things will get better"
- "I'm here if you need someone"
- "Stay strong, you can do this"

✗ Non-Hope Examples

- "There is no hope left"
- "Just smile and stop complaining" (toxic positivity)
- "Nobody cares about your problems"
- "Others have it worse; cheer up!"

Why is Hope Speech Detection Challenging? 🤖

- **Subtle expressions:** "I'm here if you need someone" has no explicit "hope" word
 - **Negation patterns:** "I have hope" vs "I have no hope left"
 - **Code-mixing:** "entha feel aayi but still I know things will improve"
 - **Script variation:** Tamil/Malayalam in native script vs romanized
 - **Low-resource languages:** Limited labeled datasets for Tamil & Malayalam
 - **Toxic positivity:** Sounds positive but invalidates feelings
- Key Insight:** Hope speech is NOT just the opposite of hate speech!

Project Goals

- Detect hope speech in **English, Tamil, and Malayalam**
- Handle **code-mixing** and **script variation** (native/romanized)
- Address **low-resource language** challenges
- Distinguish hope from **toxic positivity**
- Build interpretable baseline → scale to transformers

Datasets Used

1. HopeEDI Dataset

- English samples
- Tamil samples
- Malayalam samples
- Labels: hope, non-hope, not-in-language
- Issue:** Highly imbalanced (more hope than non-hope)

2. Jigsaw Toxic Positivity

- English only
- Emotionally supportive comments
- Discouraging comments
- Toxic positivity examples

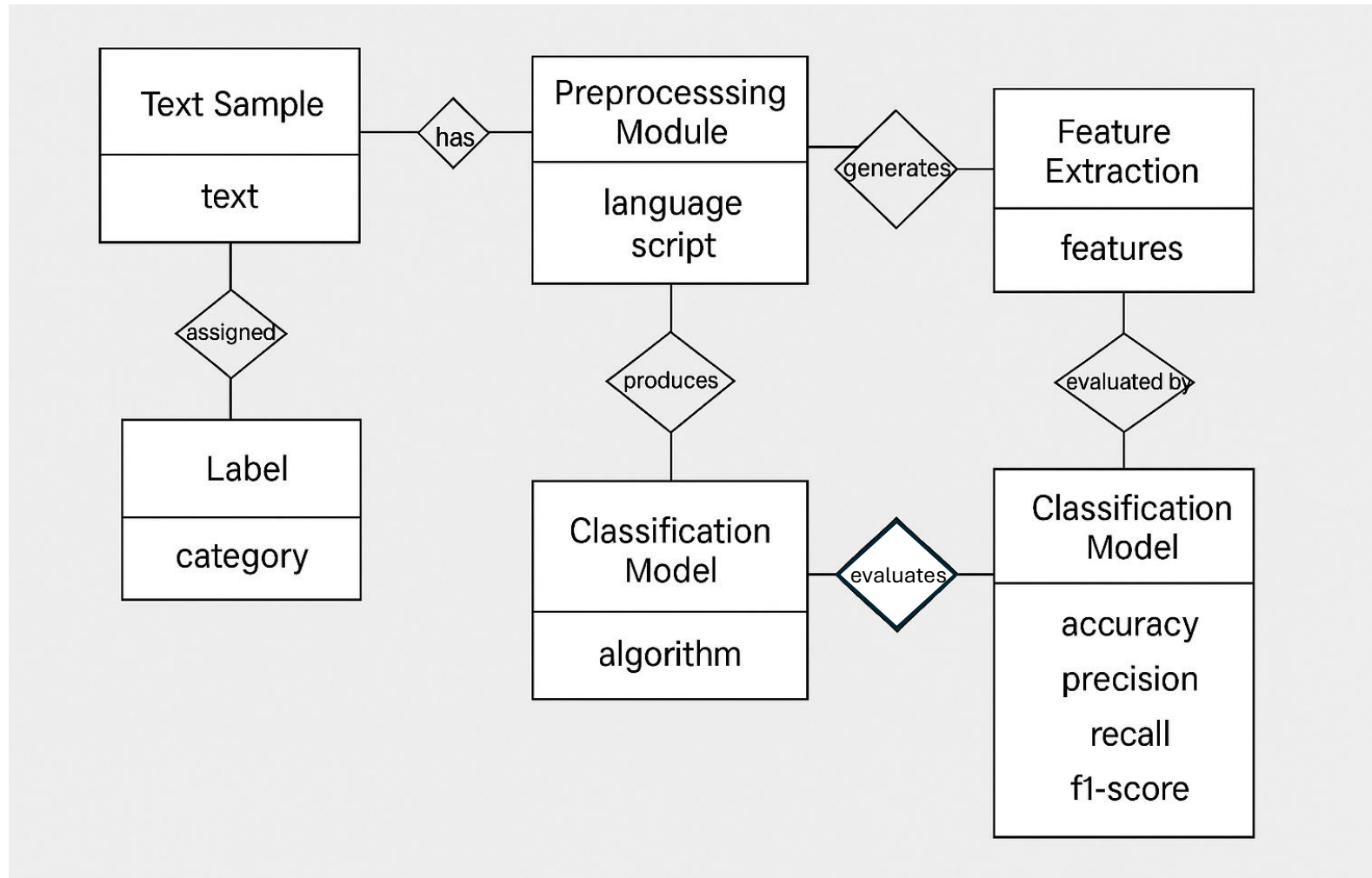
3. YouTube Comments (Exploratory)

Manually collected from Telugu, Tamil, Malayalam videos: motivational talks, devotional content, mental health awareness channels

Next Semester: Large-scale automated scraping planned

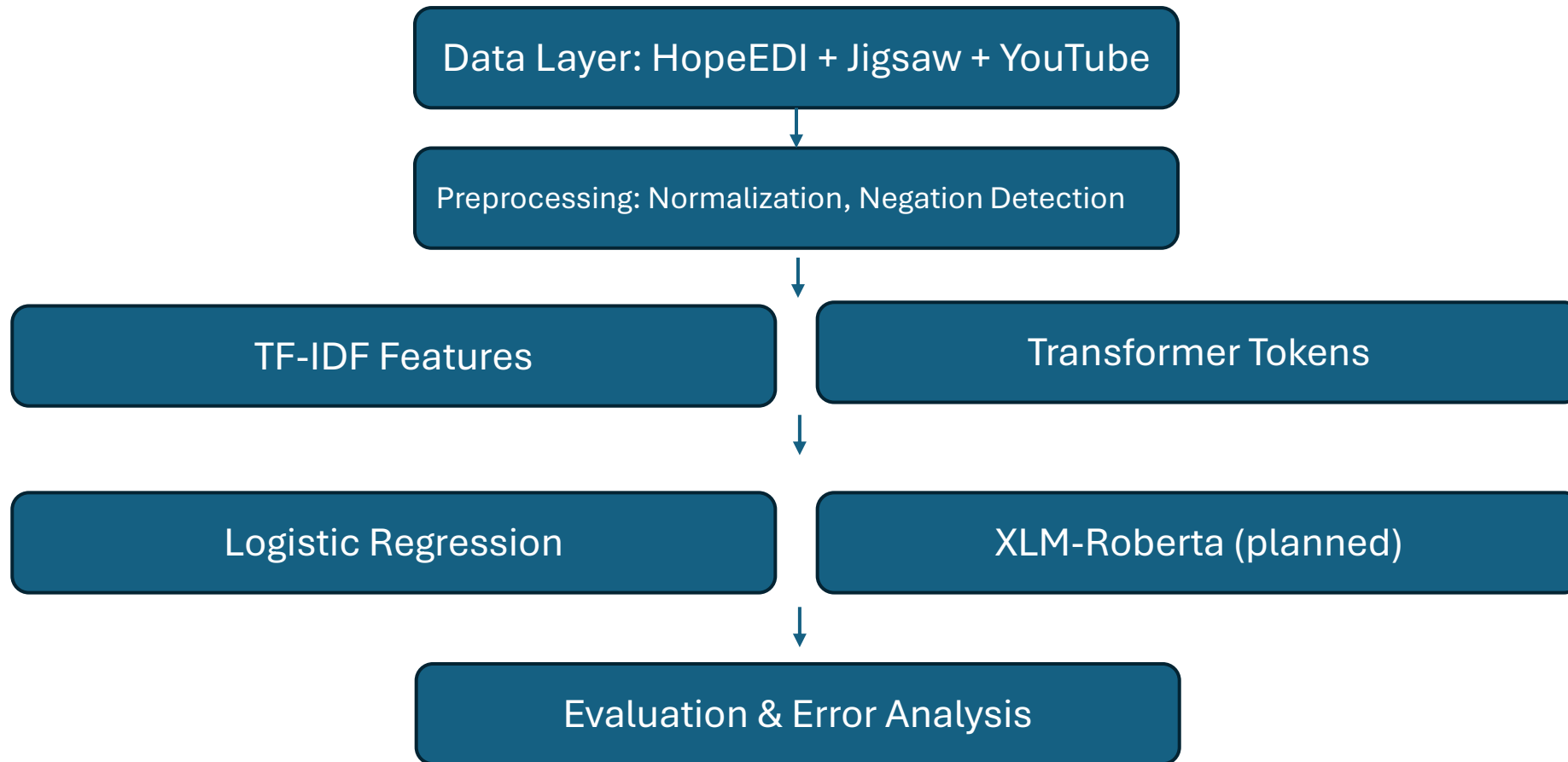
Challenge: Class imbalance - HopeEDI heavily skewed toward hope speech

Domain model Overview



The domain model defines how Text Samples flow through Preprocessing and Feature Extraction into the Classification Model, which produces Labels evaluated using standard Performance Metrics.

System Architecture Overview



Progressive Complexity Design: Start with interpretable baselines → Scale to advanced transformers

System Architecture Overview

Completed This Semester

- Data integration layer
- Preprocessing pipeline
- TF-IDF baseline model
- Evaluation framework

Planned Next Semester

- XLM-Roberta fine-tuning
- YouTube scraping automation
- Multilingual annotation
- Real-time deployment

Code Presentation

Preprocessing Pipeline

Critical Steps

- **Unicode Normalization (NFC):** Handle visually identical characters in Tamil/Malayalam
 - **Noise Removal:** Remove URLs, HTML, repeated punctuation
 - **Emoji Preservation:** Keep emojis (they encode emotional intensity!)
 - **Script Variant Tagging:** Classify as native/romanized/code-mixed
 - **Negation Detection:** Flag sentences with negation + hope words
 - **Label Normalization:** Map to integers (hope=1, non-hope=0, not-in-lang=2)
- Special Handling:** If text contains "hope" word + negation → Override to NON-HOPE

Baseline Model: TF-IDF + Logistic Regression

Implementation Details

COMPLETED

- Features: TF-IDF with unigrams, bigrams, trigrams
- Max vocabulary: 5000 features
- Classifier: Logistic Regression with class-weight balancing
- Library: scikit-learn
- Sparse matrix output

Transformer Tokenization

PLANNED

- XLM-R tokenizer
- Subword tokenization
- Max length: 128
- Multilingual embeddings

Purpose

- ✓ Establish interpretable reference model
- ✓ Create performance lower bound for transformer comparison

Baseline Results 📊

Language	Accuracy	Precision	Recall	F1-Score	Assessment
English	80.4%	91.8%	80.4%	84.2%	✅ Good baseline
Tamil	63.4%	63.3%	63.4%	63.3%	⚠️ Moderate
Malayalam	30.3%	82.3%	30.3%	38.0%	❌ Poor - needs transformers

Observation: English performs well, but Tamil and Malayalam struggle significantly with TF-IDF baseline

Error Analysis & Key Findings 🔍

Common Failure Patterns

- **Toxic positivity misclassification:** "Just smile and stop being weak" → predicted as HOPE
 - **Code-mixing challenges:** TF-IDF treats English + Tamil as separate bags of words
 - **Sarcasm:** "Oh great, another failure, what a surprise" → misclassified
 - **Cultural expressions:** Tamil/Malayalam idioms not captured by lexical features
- Why Malayalam Failed:** Complex morphology + script diversity + class imbalance + sparse data = poor generalization

Why TF-IDF Fails in Multilingual Settings 💔

- Sparse Vocabulary Explosion:** Tamil/Malayalam create thousands of unique tokens, each seen only a few times
 - Romanization Variability:** Same word appears as: "nandhi", "nandi", "nandii", "nandee" → all treated as different!
 - Code-switching ignorance:** Cannot understand context across language boundaries within single sentence
 - No long-range dependencies:** "I thought things would improve, but not anymore" → negation too far from "improve"
 - No semantic understanding:** "I'm here if you need someone" has no explicit positive words but is hopeful
- Solution:** Need contextual embeddings from multilingual transformers!

Unique Linguistic Challenges of Hope Speech 🗣️

- **Implicitness:** Hope often expressed without explicit positive keywords
 - Example: "I'm here if you need someone" → no "hope" word
- **Future-oriented semantics:** "Things will get better" requires temporal understanding
- **Cultural variation:** Tamil/Malayalam optimism encoded through cultural idioms
- **Honorifics matter:** South Indian languages use respectful forms to convey support
- **Context-dependent:** Same phrase can be hopeful or sarcastic based on context

Hope speech ≠ Sentiment analysis ≠ Opposite of hate speech

Next Phase: Multilingual Transformers 🚀

XLM-Roberta Fine-tuning (Planned for Next Semester)

Why XLM-R?

- Pre-trained on 100 languages
- Shares subword representations across scripts
- Handles code-mixing naturally
- Captures long-range dependencies
- Understands context & semantics

Implementation Plan

- Model: xlm-roberta-base
- Learning rate: $2e-5$
- Batch size: 16
- Epochs: 3
- Library: HuggingFace Transformers
- Expected: Significant improvement over baseline

Future Work (Next Semester) 🧠

1. Large-Scale YouTube Scraping

Automated extraction of 10k+ comments in Telugu, Tamil, Malayalam from motivational videos, mental health channels, devotional content

2. Multilingual Annotation Guidelines

Consistent labeling framework for hope, non-hope, toxic positivity across languages

3. Transformer Fine-tuning

XLNet / mBERT training on combined datasets

4. Toxic Positivity Dataset Expansion

Especially for Tamil and Malayalam (currently underrepresented)

Key Accomplishments

- ✓ Built complete preprocessing pipeline for multilingual hope speech
- ✓ Established baseline performance metrics (English: 80.4%, Tamil: 63.4%, Malayalam: 30.3%)
- ✓ Identified critical challenges: negation, code-mixing, toxic positivity
- ✓ Validated need for transformer-based approaches

Broader Impact

Enables positive content moderation, mental health support systems, and community resilience monitoring for **non-English speaking populations**

Code Run